Shadows of Almia Showdown

Project brief and current state of the work, for handing the context to another assistant. Written 27 October 2026.

1. What the project is

Shadows of Almia Showdown is a browser Pokémon battle royale built by Wesley Millet with Claude (Anthropic) as the engineer. Eight trainers land on a 320 x 224 tile island, catch and train wild Pokémon, fight one another in one-on-one trainer battles, and the last trainer standing wins while a creeping darkness shrinks the safe ground toward the Shadow Crystal at the centre. It is a non-commercial fan game: Pokémon sprites come from the PMD SpriteCollab project, the human sprites from Pokémon Ranger: Shadows of Almia, move effects from Pokémon Ranger: Guardian Signs, and the island is painted with The Fan-tasy Tileset (a purchased pixel-art pack by Valerio Colonna). Keep it free, no ads, no payments.

Where	Address / location
Live online server (Fly.io)	https://almia-showdown.fly.dev
Offline demo (GitHub Pages, branch main)	https://wesmillet.github.io/Shadows-of-Almia-Showdown/
Repository	github.com/wesmillet/Shadows-of-Almia-Showdown
Branches	**online**: the server plus the online client, deployed to Fly. **main**: the offline demo (no server/ folder, no online UI), deployed to GitHub Pages. Every change is committed on online and cherry-picked to main (server-only files dropped).
Deploying	GitHub Actions workflow "Launch online server" (launch-online.yml) with ref main and input size=friends deploys the live server; "smoke-online.yml" runs an end-to-end smoke test and writes the result to a GitHub issue. Pushes to main also rebuild the Pages demo.

2. Technology and architecture

- **Client**: TypeScript, Vite, one HTML entry (index.html). Canvas 2D draws the world; a hand-written WebGL pass does lighting. No game framework. Character sprites draw at an integer 3x with nearest filtering; the ground scales with the window.
- **Simulation** (src/sim): a deterministic 60 Hz fixed-step match (match.ts), battle rules (battle.ts), AI trainers (ai.ts), catching, items, evolution, darkness, fishing, mega evolution, visitors (merchant, Ace Trainer Celeste, Hoopa, Lugia), townsfolk, seats for up to eight humans. The same code runs offline in the browser and on the server.
- **Server** (server/, branch online): Node 22, TypeScript via tsx/esbuild, the ws WebSocket library, SQLite through node:sqlite. One program serves the pages and art, runs every match in a worker thread (remoteRoom.ts, roomWorker.ts), the matchmaking lobby, global chat, accounts and saves. Browsers send only keys, cursor and commands; 20 times a second each player receives the changes to the part of the island they can see (view.ts, src/net/sync.ts) and mirrors the match (src/net/mirror.ts) with client-side prediction for movement.
- **Data** (data/*.json): species (322, with learnsets, base stats, sizes), moves (175, each with an animation recipe), items (64) plus TMs and loot tables, types, spawn tables per land, characters (who plays which NPC role), fishing, shiny palettes, and config.json (every tunable: speeds, darkness schedule, catch rates, outposts, respawn, lighting, sizes).

- **Assets** (assets/): 324 Pokémon sprite folders (SpriteCollab AnimData.xml + sheets), portraits, ranger sheets with a manifest, 256 Guardian Signs effects, item icons, fonts, music, UI art, and the baked island (assets/maps/almia3.json, a tile atlas, object sheets, minimap, the pixel paving overlay and the colour blend image).
- **Map generator** (scripts/fantasy/*.py, Python with numpy, scipy, Pillow): designs and paints the island from the Fan-tasy tileset and exports it. Described in section 5.
- **Tests**: Vitest unit tests (81 on online, 56 on main), a headless whole-match run (server/headless.ts), end-to-end server checks (server/checks/*.ts: social, teams, lag compensation, hitches, moderation, ops, launch_check), CI on every push (.github/workflows/ci.yml).
- Size: about 42,000 lines of TypeScript and Python, 294 commits.

3. Repository layout

Path	What lives there
src/core	fixed-step loop, input, rng, noise, math
src/sprites	SpriteCollab AnimData and Offsets parsers, sprite loader, ranger sheets, item icons
src/ground	GroundRenderer interface; fanGround.ts bakes the Fan-tasy map in chunks (ground layers, animated tiles, the paving overlay, the colour blend)
src/render	camera, world drawing (depth-sorted objects), WebGL lighting
src/sim	the whole game simulation (see above)
src/ui	title, starter roll, match screen and HUD, end screen, PC, trade menu, social panel, online screens (hub and lobby), options, move gallery, watch screens
src/net	protocol, sync encoder/decoder, mirror, connection, progression (trainer XP, looks, achievements)
src/vfx, src/audio	move effects library and visuals, floating text, music
server/	hub, matchmaker, room and worker threads, accounts (SQLite), chat, moderation and mod dashboard, rating (Glicko-2), replays, backups and S3, telemetry, events, profile and privacy pages, admin CLI, bots, checks
scripts/fantasy	the island generator (design3, paint3, cliffs, castlekit, castle3, farm3, content3, decor3, polish3, animprops, areas, towns3, export3, tiled, wang, automap, statue)
scripts/mapgen	post-render passes: pave_smooth.py (pixel-smoothed paving), center_blend.py (biome gradient and the grey centre), minimap_from_render.py
scripts/*.py	data builders: fetch-sprites.py, learnsets.py, moves.py, megas.py, shiny-palettes.py, add_starters.py
tools/mapshot.html + src/tools/mapshot.ts	renders the island in the real engine for review (parameters: layer=ground|decor|all, blend, overlay, skip)
docs/	online.md (how online play, deployment, moderation and backups work), launch-day.md, plan.md
assets/maps	the exported island; assets/ui/icon the browser icon (a pixel Lugia head)

4. Gameplay as it stands

- **Start**: title screen (a live corner of the island), roll three starters by rarity (Common 50%, Uncommon 27%, Rare 13%, Epic 7%, Mythic 3%), pick one, land at a random coastal spot with seven rivals spread round the coast.
- **The island**: the Shadow Crystal on a paved oval battlefield at the centre of a blighted plain (grass drained ash-grey, dead trees, bones, glowing rune rocks); four lands round it, each with its own wild table, a boss and a

town: Frostmere (snow, on the foothill of the ice mountain), Emberwood (autumn), Sunreach (desert), Greenhollow (forest). The castle stands in a clearing in the south-east forest. Tide Lake (Lugia) lies between the plain and the autumn wood. Roads wind between everything and climb plateaus on stairs.

- **Fighting**: two Pokémon out at a time, moves on cooldowns aimed with the mouse, auto-battle (R) for the lead, a dodge, a Capture Styler loop (hold C, now a rare held item that shows a spinning cursor icon), balls that lock on with F. Trainers within six tiles start a one-on-one battle in a ring; the loser is out. Statuses: burn, paralysis, sleep, poison, freeze, confusion.
- **Progress inside a match**: XP and levels, evolution at canonical levels, stones, TMs, held items, Mega Evolution (ten species), Elite, Alpha and shiny wilds, money and shops, fishing with three rods, flying on Latios, respawning at an outpost after a wild knockout (with a growing wait), Extraction Vouchers to send a Pokémon to the PC mid-match.
- **Healing**: a free healer in every town and in the castle courtyard; a paid Crystal Healer (P$600) on the plain south of the arena, closed during the final.
- **Visitors and legends**: the Wandering Merchant, Ace Trainer Celeste, Hoopa (one match in seven; three hits enrage it; opens a portal to a legend), Lugia hooked from Tide Lake with a Super Rod, Darkrai at the end. Hoopa, Shadow Lugia and Shadow Mewtwo are kept secret in marketing. Camera shake only for Lugia, Darkrai and Shadow Mewtwo.
- **The end**: every 200 s the crystal pulses and darkness creeps inward; when two trainers remain they are called to the oval for the final battle.
- **Between matches**: a PC box (10-slot storage chest, Pokémon kept after a win or by voucher), a trainer card with stats (Wins, Eliminations, Matches, In PC), trading and Wonder Trade between accounts, global chat on the right of the hub with clickable names, a profile page per player (/u/name), replays.

5. The island and its generator

The island is produced by a Python pipeline from The Fan-tasy Tileset (the pack lives outside the repo; set FANTASY_DIR to its folder, the one holding Art/ and Tiled/). The pipeline, in order (a rebuild script runs all of it):

- **design3.py**: the shape of the land (coast, beach, four lands by angle, the central plain), the plateaus and the ice mountain (MOUNTAIN: five lobed tiers over the foothill, set back seven rows at the south faces and four at the north, faces squared where the trail climbs), lakes, Tide Lake, the river from Tide Lake to the north coast, the stream and waterfall on the foothill, roads (winding centre lines through waypoints; climbing roads run straight through their stairs), the castle rectangle and gates, towns, STAIR_SITES, SUMMIT, BLIGHT.
- **build3.py**: paints terrain with the pack's terrain sets (wang.py reproduces Tiled's terrain brush), cliffs with the pack's own automap rules (automap.py, cliffs.py; ice slopes on the snow mountain, grey and brown slopes elsewhere), roads, tall grass, the castle area, farms, town enclosures, bridges, the animated-props layer, woods and scatter, the blight, the mountain decor, the desert ruins, shores, life (flower patches, reeds, lily pads, logs, braziers, torches), collisions from the pack's collider shapes, walkable and reachable grids, then exports.
- **castlekit.py** + **castle3.py**: the castle is composed piece by piece from the Atlas_Castle_Building kit (walkway ring painted with the Wood Floor terrain, parapets, faces, bases, round corner towers, the gatehouse arch, red-roofed towers at the side gates, the keep with its roof, crenellated band, door and stairs), with a garden courtyard (kerbed lawns from Tileset_Castle_Grass, a paved plaza, the battlefield oval, walks, a fountain with a stone Celebi statue, trees, hedges, flowers, benches, lamps, a training yard, stores, a camp). Nothing is cut from a sample map any more.
- **areas.py**: the castle is also a Tiled map (scripts/fantasy/areas/castle.tmx) that references all 131 tilesets of the pack. build3 --export-castle writes it; areas.py install/fetch copy it to and from the pack's Tilemaps folder; when the file exists the build stamps it in (tile layers as drawn, a Solid layer for walls, Paving shapes that get pixel-smoothed, Objects, Buildings as image layers with a foot property, Markers for the healer, challenger,

statue and arena). The round trip is exact. areas/README.md and the project doc "Editing the castle in Tiled" explain it.

- **polish3.py**: a landscaper pass (groves, flower drifts along roads, beach life, reeds, open-field singles) that keeps off the castle, the arena oval and the blight.
- **Render passes**: tools/mapshot.html renders the island in the real engine; scripts/mapgen/pave_smooth.py redraws paving edges at pixel level (true ovals, rounded corners, a rim, worn turf) into assets/maps/almia3_overlay.png; center_blend.py writes a per-pixel colour multiplier (assets/maps/almia3_blend.png) for the gradients where the plain meets snow, desert and autumn and for the ash-grey centre (grass only, tall grass keeps some green); minimap_from_render.py takes the minimap and preview from the final render.
- **Runtime**: src/ground/fanGround.ts bakes the stack of tile layers per chunk, animates tiles (the Animations layer holds torches, campfires, braziers, banners, flags, flowers, water plants, the waterfall, rune rocks), draws the paving overlay under the Animations layer, then applies the blend. Objects (trees, props, buildings, statues) are depth-sorted pictures.

Tileset use after this session: about 70 of the pack's 130 tilesets (up from 36): ground, sand, water, snow, leaves, road, tall grass (all five kinds), rock slopes (ice, grey, grey-sand, brown-snow), castle building, floor and details, castle grass, farm field, fence 1 and 2 (plus snow), wall 1, every seasonal tree, bush, rock and prop set, grass clumps, the snow buildings (tower, watchtower, shelters), bridges, and the animation sets (torch, campfire, brazier, banner 1 and 3, flowers, water plants, water props, waterfall, rune rocks). Still unused on purpose: the build-your-own house atlases (towns use the artist's finished houses), the city-wall set (its automap rules are incomplete and reference a renamed tileset), the sky backgrounds.

The island as it is now (full render 5120 x 3584 px, one pixel per game pixel):





The ice mountain: five tiers, the trail on stairs with torches, the summit tower, camps.



The castle composed from the kit, with its garden courtyard and the battlefield oval.

6. Online play, operations and safety (branch online)

- **Modes**: Casual (anyone, alone or in a party of up to four playing as a squad), Ranked (signed-in, solo, Glicko-2 extended to free-for-all placements, five placement games, tiers Bronze to Master, 60-day seasons), Private lobbies (five-letter code; rules for trainer count, bots, starter level, teams, storm speed, rare chest; nothing kept), Friends, Watching (friends, the featured match, the rest of your own match; spectators run 4 s behind), Replays (every match, keyframed, oldest dropped past REPLAY_MB).
- **Lobbies**: a match needs at least half a lobby of humans before bots fill the rest (MIN_PLAYERS, default ceil(size/2)); the lobby shows "Waiting for N more trainers"; "Start now" is ignored below the minimum.
- **Accounts**: trainer name and password (scrypt), optional Sign in with Google (needs GOOGLE_CLIENT_ID/SECRET), eight one-time recovery codes instead of email, sessions as HttpOnly cookies. Saves: PC box, storage chest (10 slots), look, stats, achievements, titles. Trainer levels exist in code but are switched off (LEVELS_ON = false in src/net/progression.ts).
- **Moderation and anti-cheat**: the server runs every match, so a modified browser cannot cheat; impossible commands are refused and flagged, enough flags file a report. Chat filter, strikes, mutes, reports, a moderator dashboard at /mod, an admin CLI (dist-server/admin.js).
- **Operations**: Fly.io app almia-showdown (shared-cpu-2x, 1 GB, about six matches; scale up, never out: one machine, the lobby in memory, SQLite single writer). Each match in its own worker thread. Backups every 15 minutes to the volume and, with Tigris S3 set up, off the machine. Alerts to a webhook (ALERT_WEBHOOK_URL). Metrics to Fly Grafana. Gentle deploys drain running matches (OPS_TOKEN). Health at /health. GoatCounter for page views and match events.
- **Not yet done on the ops side**: a real domain with HTTPS certificates (PUBLIC_URL), sizing the machine for a launch (performance-1x or better), confirming off-site backups and the Google sign-in secrets on the live app, a Discord alert webhook, phone controls, the "first tab closed with code 4000" timing message in the smoke test, which is a known cosmetic failure.

7. What was done in the most recent session (27 October 2026)

- Global chat moved to the right of the hub; trading between accounts, Wonder Trade, clickable chat names opening a trainer card with a Trade button; an Eliminations stat.
- PC sizing and centring fixed for every sprite, including the hover animation.
- A paid Crystal Healer near the arena; the healing outposts restyled (a nurse sprite recoloured from a townsfolk sheet, a white flag with a red cross).
- Hoopa enrages after three hits; a spawn-rate study (241 forms, about 218 wilds per match); Hippowdon replaced by Hippopotas; lists of candidate desert and snow additions were given but not yet chosen.
- Gradient transitions between the plain and the three bordering lands; then the plain's centre drained to ash grey with dead trees, stumps, logs, skulls, broken tombstones and rune rocks, confined to the grey ground.
- The ice plateau given more exits, the desert outpost moved north, roads widened (never one or two tiles), roads converging on the arena at north, south, east and west; the snow roads wind again.
- The castle rebuilt several times on feedback: first a stretch of the pack's Bosk Barracks sample, then a garden courtyard, then composed from the castle kit itself; the battlefield oval and the fountain with a stone Celebi kept throughout; the castle exported as an editable Tiled map.

- Chests cut to 10 slots; the dev chest only with ?rare=1; trainer levels switched off; bots only once half a lobby is human; the "empty seats get AI trainers" line removed; the online menu got a map backdrop; camera shake limited to the three legends; the Capture Styler made a rare held item with a cursor icon; a stuck-after-reload bug fixed (undefined fields now travel as deletions in the sync protocol); the browser icon set to Wesley's pixel Lugia head.
- The island rebuilt with far more of the pack: the ice mountain in the style of the pack's Frost Peak (tiers, stairs, torches, summit tower, braziers, banners, camps, watchtower, flags, waterfall), farms by Greenhollow and Emberwood (fields, crops, fences, farmhouse, haystacks, cart, well, scarecrow), a river with a stone bridge, town enclosures (stone walls or fences with gaps for roads, trees, a watchtower and props inside), the desert ruins, boats and palms on the shores, torches at town plazas, animated life everywhere, denser woods.
- Every change was deployed to the live server and ported to the offline demo; all tests pass on both branches.

8. Conventions to keep

- Never use em dashes in any writing (code comments, commit messages, documents, chat).
- Commit messages end with two lines: "Co-Authored-By: Claude Fable 5.1 <noreply@anthropic.com>" and "Claude-Session: https://claude.ai/code/session_019L7iVyevzrSsZHNd1CijFN".
- Work on branch online, then port to main (cherry-pick; drop server/ and online-only files on conflict), then merge main back into online with the ours strategy so the branches stay related.
- Keep Hoopa, Shadow Lugia and Shadow Mewtwo out of marketing material.
- Secrets: a Fly.io org token and 2FA recovery codes were once pasted in chat; they were to be deleted and regenerated and must never be reused or stored.
- Build from the pack's kits and individual assets, not by cutting pieces out of its sample maps. Use the terrain sets and automap rules the artist provides. Every tile and object must exist in the pack.
- The map build needs FANTASY_DIR pointing at the pack folder; the rebuild order is build3, polish3 --write, a ground render, pave_smooth, a ground render with the overlay, center_blend, a full render, minimap_from_render.

9. Open items and likely next steps

- Choose the desert and snow spawn additions (ready now: Snorunt, Swinub, Bergmite, Smoochum, Lapras; Sandshrew, Cacnea, Geodude, Gible; others need sprites fetched).
- Bring more of the centre arena to life with the pack's assets (the user asked for a cooler arena; braziers and rune rocks were added, more could follow).
- Towns built from the build-your-own house atlases (a house grammar from the Atlas_Buildings kits), the remaining pack sets (city walls, sky backgrounds, 1-tile rock slopes).
- Operations for a public launch: domain and certificates, machine size, backups off the machine, Google sign-in secrets, alert webhook, phone controls, a fix for the two-tab 4000 message in the smoke test.
- Make towns and the arena editable areas in Tiled the way the castle is.

10. Key numbers

Thing	Value
Island	320 x 224 tiles of 16 px (5120 x 3584 px); 25 ground layers; 2,172 placed objects; 18 NPC spots; 5 healing outposts; 4 towns

Thing	Value
Content	322 species (324 sprite folders), 175 moves, 64 items plus 38 TMs, 18 types
Code	about 42,000 lines (TypeScript and Python), 294 commits; 81 unit tests on online, 56 on main
Server	Fly.io shared-cpu-2x 1 GB; one match about 9% of a core and 2 KB/s per player; 8 players per match, 60 s lobby wait live, MIN_PLAYERS half the lobby
Pack	The Fan-tasy Tileset: 130 tilesets, 4,088 art files, 16 sample maps, documentation PDF; about 70 tilesets now in use